



Himalayan Bamboo Pvt Ltd

Rebuilding a Sustainable Nepal

In association with



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Bamboo can be a viable housing option

- ✿ Strength
- ✿ Higher Earthquake Resistance
- ✿ Energy Efficiency
- ✿ Durable with proper treatment
- ✿ Grows widely, denser, faster
- ✿ Potential carbon credit
- ✿ Renewed Local livelihoods (supply chain)

How is bamboo house earthquake resistant?

- **Bamboo structure gives:**
 - Higher tensile strength than many alloys of steel
 - Higher compressive strength to concrete mixtures
 - Higher strength to weight ratio
- **Single storey:**
 - Light weight
- **Symmetrical layout:**
 - Evenly distribution of load
- **Interconnected walls:**
 - Whole structure acts as a single entity

Bamboo as a green material

- ✿ Has a highest growth rate – up to 90 cm a day
- ✿ Has a short rotation – 3-6 years
- ✿ Can grow densely (5-10,000 culms/ha)
- ✿ Has a natural regeneration capacity
- ✿ Grows in different climates and poor soil condition

Bamboo as a green material

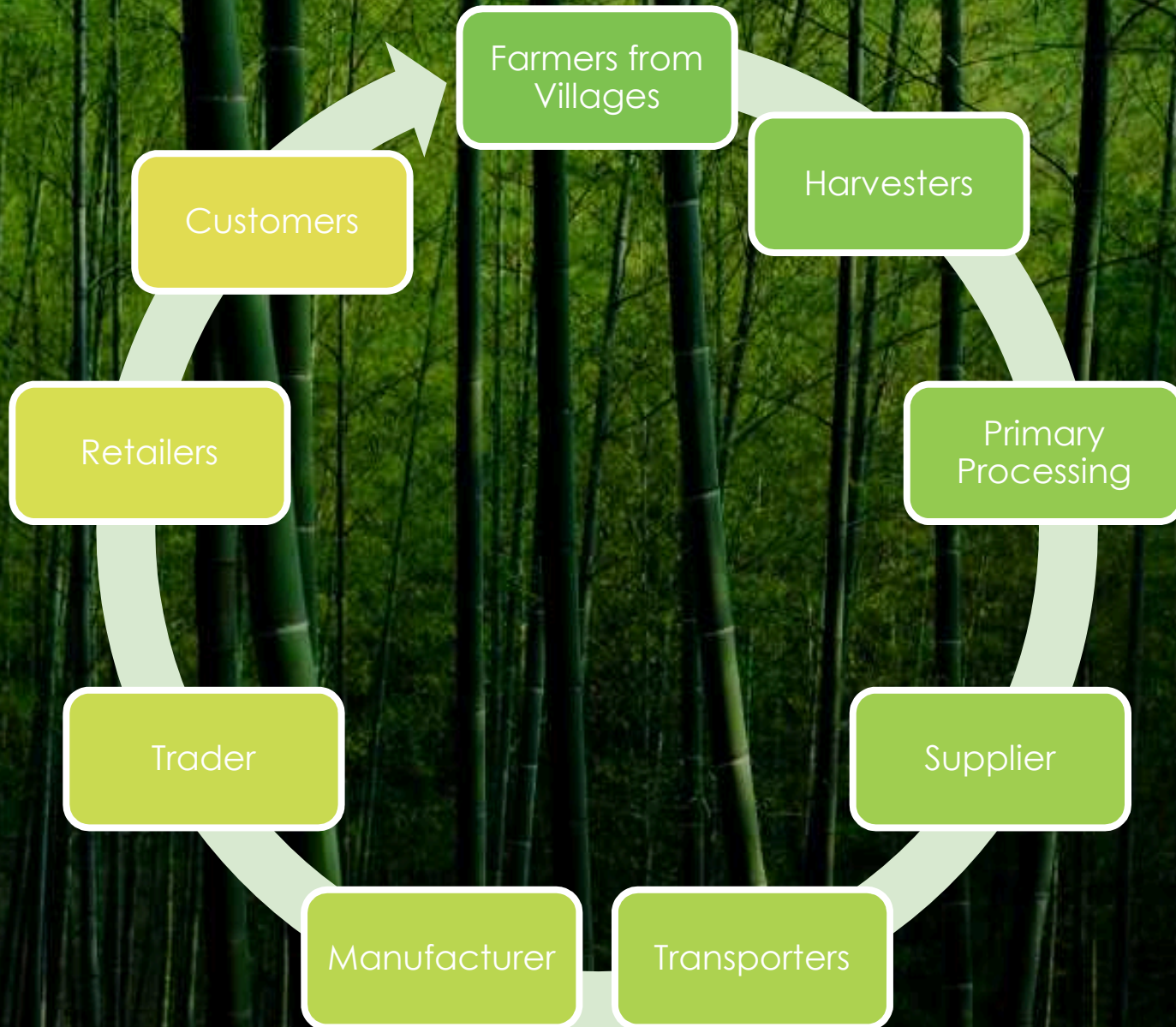
- ❁ **Energy efficient processing** – Studies show that processing of bamboo requires only 1/8th the energy for processing of concrete and 1/3rd of that of wood to create a building material of the same capacity. In comparison to steel, bamboo needs only 1/50 of the energy for processing (Roach 1996).
- ❁ **Check deforestation: 70 hectares bamboo for 1000 houses and would be otherwise 500 hectares of trees**
- ❁ **High carbon sequestration rate (more than 30% that of trees)**

Challenges of Bamboo House

- ❁ Social Acceptance
- ❁ Government has not set any national standards for bamboo house
- ❁ Cannot be re used again
- ❁ Unsystematic use of bamboo



Bamboo Supply Chain



Major Resource Comparison (Basing 360 sqft house)

Resource	Bamboo	RCC
Cement	4000 kgs	12,500 kgs
Sand	400 cu.ft	750 cft
Aggregate	80 cu.ft	675 cft
Bamboo/Bricks	250 no.s	9200 no.s
CGI Sheet	3.33 bundle	-
Rods	-	2 tonnes

- A RCC house would cost a family approx. 8 to 12 lakhs of 360 sq.ft.
- The same house would costs approx. 5 to 8 lakhs.

Company's Objective

- ❁ To provide affordable and eco-friendly bamboo housing option.
- ❁ To uplift local income by introducing various kinds of bamboo based local enterprises, employment generation, market development and promotion of bamboo as easily saleable cash crop.
- ❁ To increase access to education and health facilities of rural communities by developing bamboo based low cost infrastructures like schools, clinics, community hall (to enhance physical assets).

Some of the Previous Projects

Project Name	Sq. Ft	Location
Camp for Muku Kamaiya	300	Kanchanpur
Residential House	580	Biratnagar
Con Tech Pvt Rims Nepal	600	Melamchi
Rims Nepal Training Hall	1000	Baireni
ECG Center Bhawani Ekikrit	528	Lahan
Bhawani Ekikrit School	880	Kasaha, Siraha
FNCCI Trade School	1313	Surkhet
FNCCI Trade School	684	Surkhet
Mudhbary & Joshi Construction Office	1200	Middle Marsyangdii
Milk Collection Center (In association with World Vision Intl Nepal)	609	Syankudi
Chandi Lower Secondary School	1800	Bharatpokhari, Kaski
Training Hall Lekhnath Chamber of Commerce	1521	Lekhnath
SOS Bahini Eco Village Pokhara	1373	Satinagar, Pokhara

Baireni, Dhading



Rotary School, Pokhara



Rotary Rehabilitation Center, Pokhara



SOS Baini Eco Village, Pokhara





Construction Process

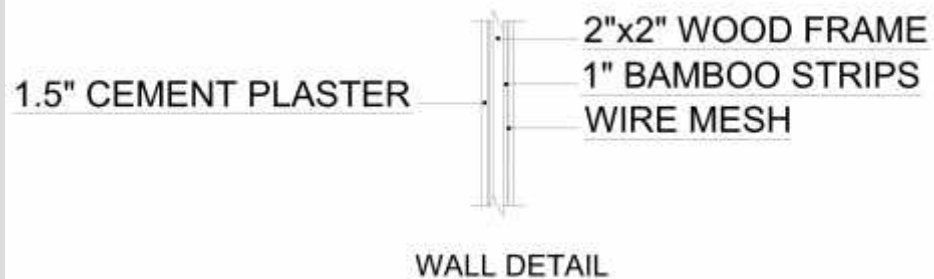
Bamboo Treatment (CCB)

- Copper Chrome Boron
- Anti Termite treatment chemical
- 6-8% for indoor use
- 8-10% of outdoor use

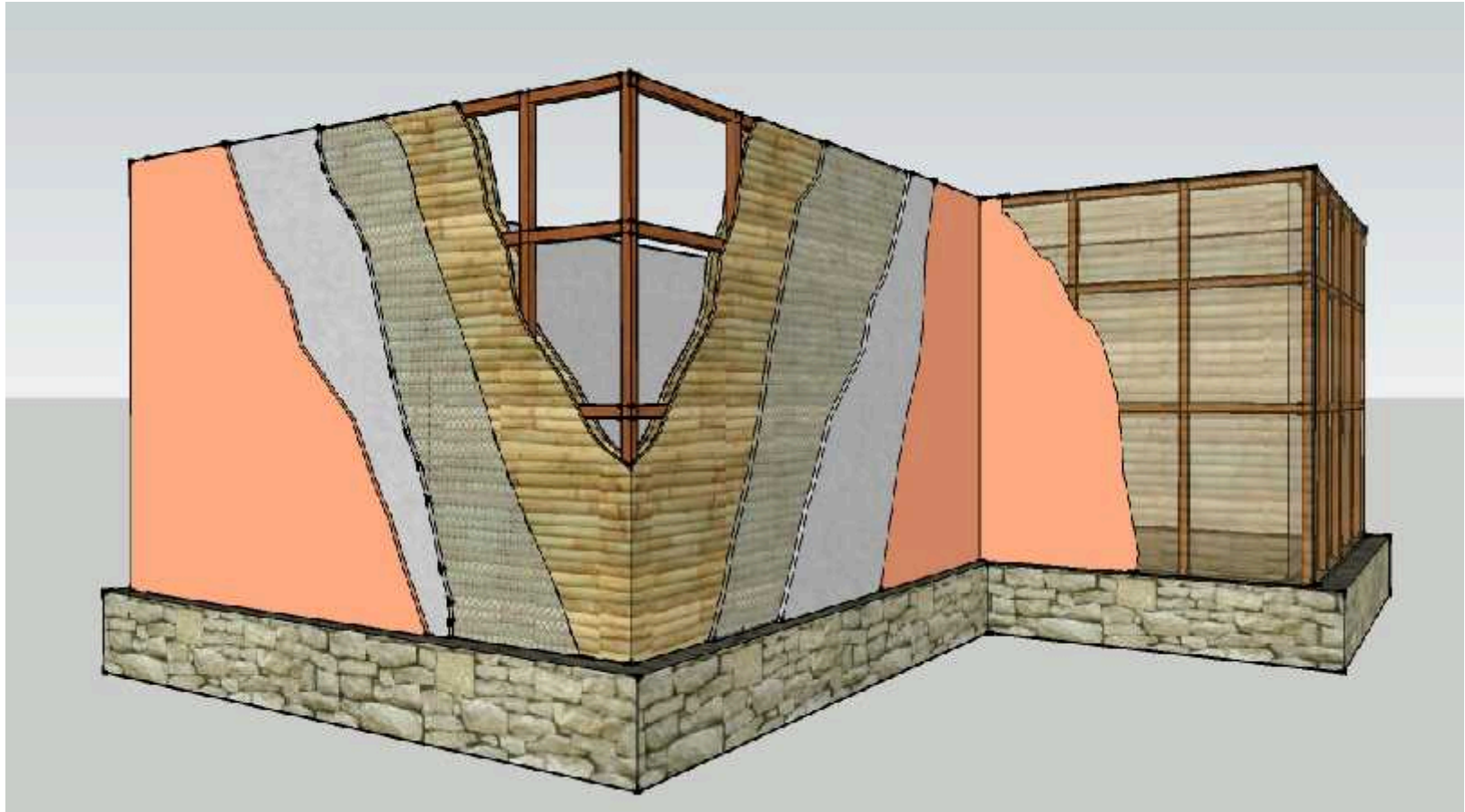


Baharaque Technique

These walls have timber frame with split bamboo in the middle covered with cement plaster on both sides



Layers of the wall





१. जग निर्माण



४. छाना लगाउने काम ।



७. रङ्ग लगाउने काम ।



२. बाँस र काठबाट निर्मित तयारी भित्ता जडान



५. सिमेन्ट लगाउने काम ।



८. बस्नको लागी तयार बाँसको घर ।



३. घरका भित्ता जडान पश्चात ट्रेस निर्माण ।



६. सिमेन्ट लगाउने काम ।



९. भुकम्प प्रतिरोधि बाँसको घर ।

Primary Plinth Of The House

Basic foundation of the house.



15 days for the plinth to be ready for other construction.



Assembly of the house

The wall assembling.



Roofing of the bamboo house

Support to the roof.



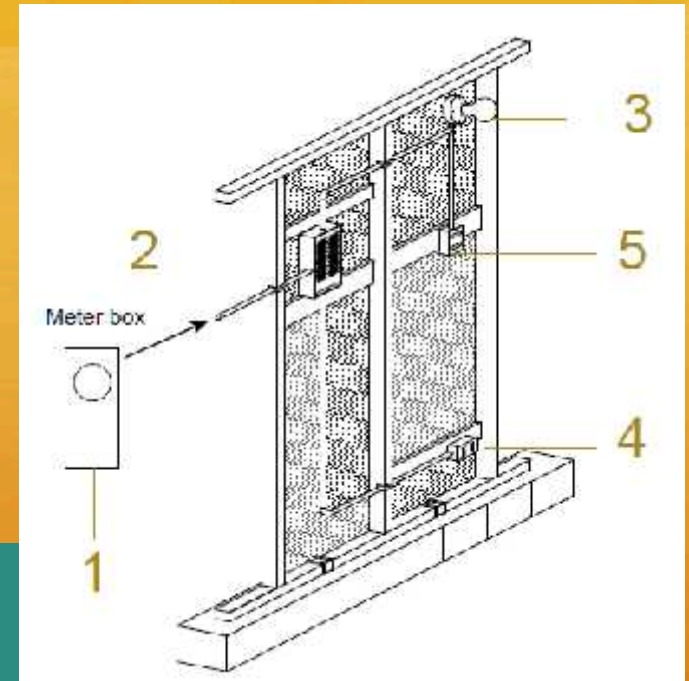
Upper View



Inner View of the Roofing



Electrical Installation



Electrical installation components:



1 the meter



2 Board general



3 lightpoint

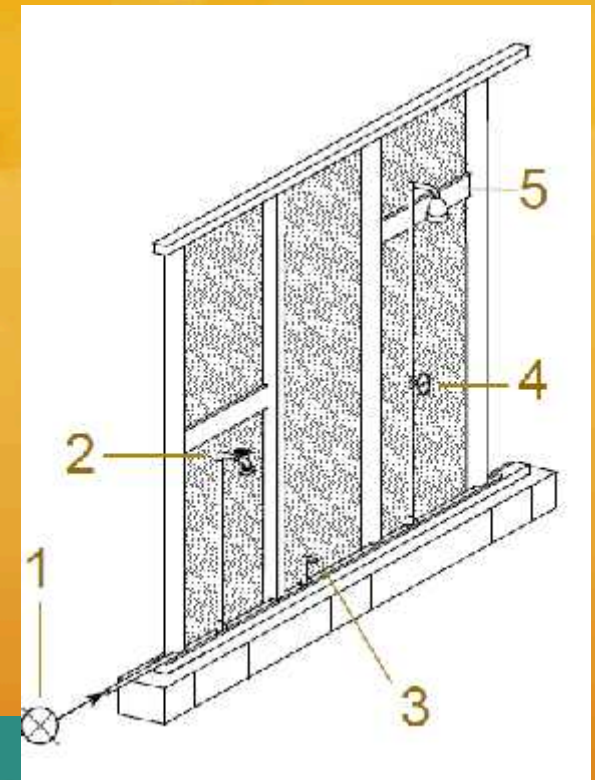


4 receptacle



5 switch

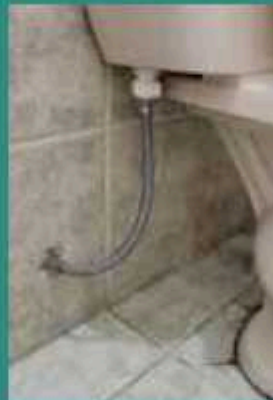
Sanitary Installation



1 Valve switch



2 Faucet sink



3 Rush bath



4 Key shower



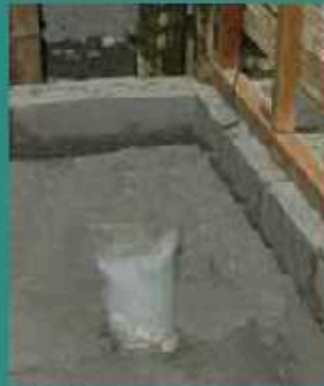
5 Shower

Drainage Installation

Drain installation components:



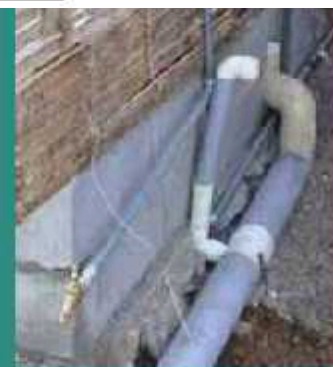
1 Laundry evacuation



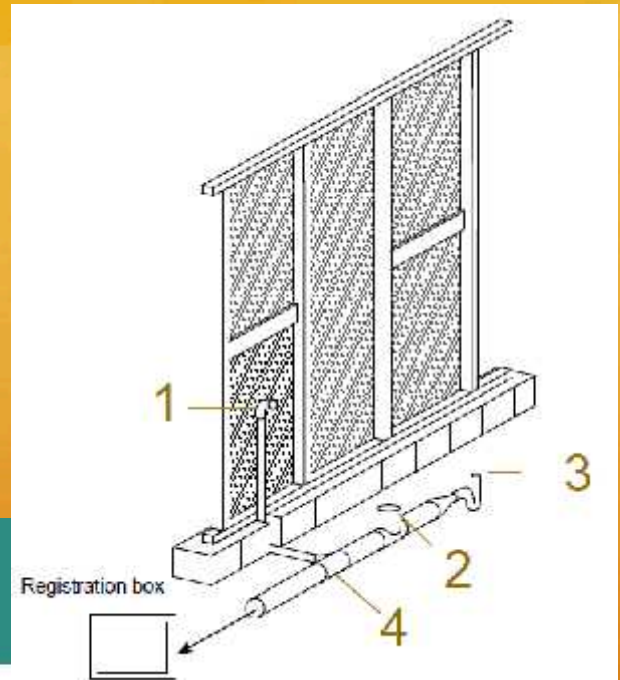
2 Toilet pipe 4 "



3 2 "pipe for evacuation shower elbow to prevent odors



4 Connection between pipes 2 "and 4"



Plastering of the bamboo House

Plastering and support



Controlling the plaster level

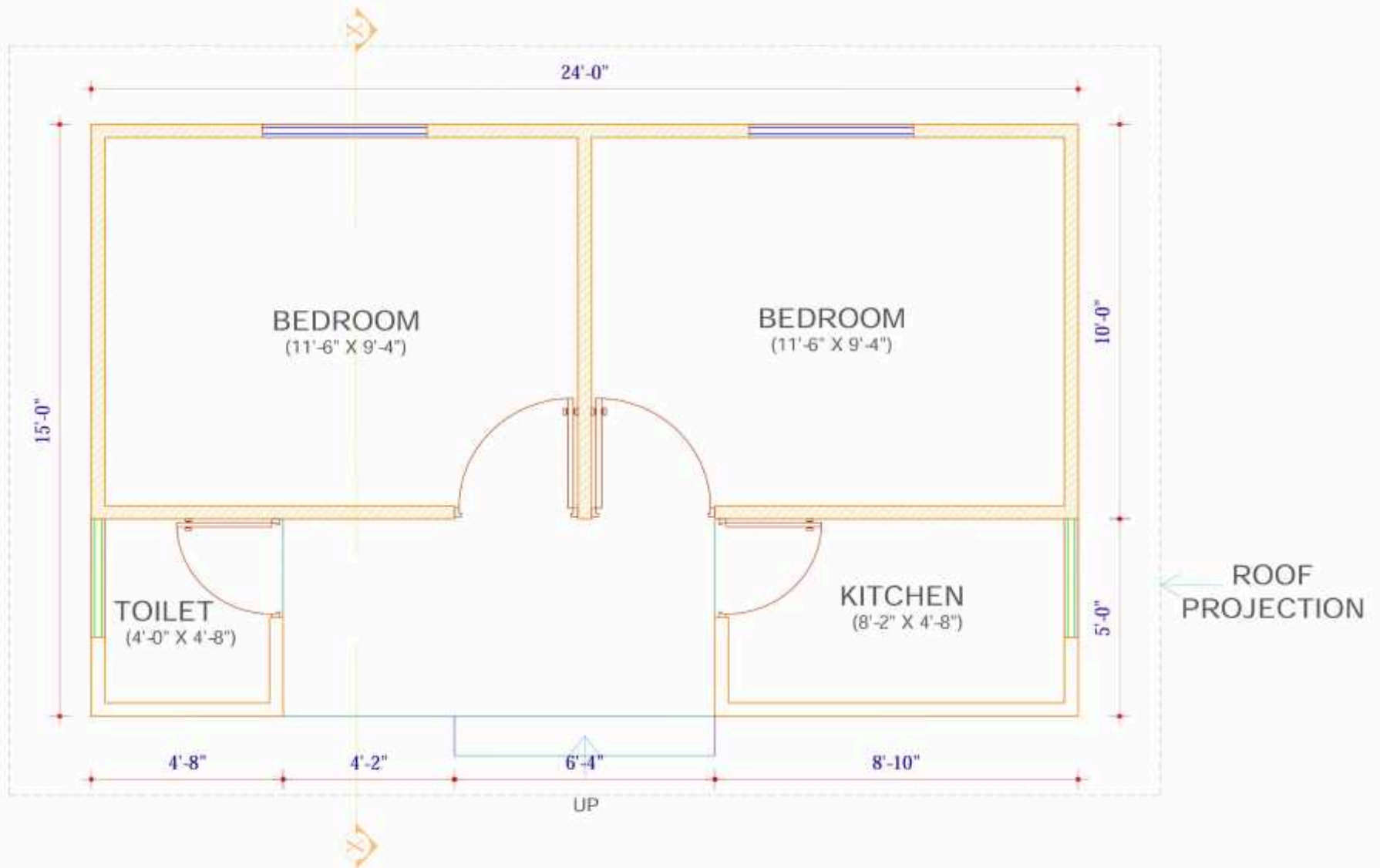


Varnishing/Painting/ Final Touch

Final House Ready



Bhaktapur Model House



FLOOR PLAN

(Coverage: 360 sq. ft.)

Type A



Thank you.